Size-Agnostic Safe Reinforcement Learning for V2G Charging with a Feasibility-Preserving Safety Layer

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[TODO: The author chooses the final title; this is the working title.]

Abstract

Charge point operators offering vehicle-to-grid (V2G) services must keep a station within its transformer limit and deliver each driver's agreed energy by departure. Reinforcement learning (RL) controllers usually treat both requirements as reward penalties, and their networks are tied to one station size. This paper combines size-agnostic set policies with a safety layer, an exact projection onto per-vehicle power bounds and the two-sided transformer limit. The lower bound of each electric vehicle (EV) keeps its admitted requirement reachable at full power. For any sequence of raw actions, the transformer limit provably holds at every step under simulator and grid-headroom conditions. Every EV whose stay contains no infeasible step, where the transformer cannot cover all lower bounds, meets its admitted requirement.

Attention and Deep Sets policies were trained with proximal policy optimization (PPO) on 10 and 20 chargers in EV2Gym and tested without retraining on up to 200 chargers. With two training seeds per learned method, no transformer overload occurred in 102,750 test episodes run through the safety layer, against 3,730 and 3,274 overload steps for penalty-trained and Lagrangian attention policies. For every non-conformal method with the known-departure bound, no EV was left short in an episode without infeasible steps.

At 200 chargers, the attention policy had a gap of 0.90 euros per charger and episode to a perfect-information optimum, against 0.87 euros when trained at 50 chargers. The Deep Sets policy reached 0.68 euros, so the attention architecture showed no advantage. A price-threshold rule with the same layer had a lower mean gap than the attention policy at every size of the test grid. The Deep Sets policy beat this rule only from 20 chargers upward.

Learned policies caused infeasible steps, and for the attention policy the share of EVs without one rose from 65.9% at 5 chargers to 92.6% at 200. With the departure bound, this policy left 2.24% of admitted EVs unserved, against 1.86% for its penalty-trained twin and 0.99% for a variant without the bound. That variant also had a lower gap at every size. In penalised profit, the attention policy scored 0.04 to 3.46 euros per admitted EV more than the twin, with an interval including zero only at 200 chargers.

Keywords: Electric vehicle charging, Safe reinforcement learning, Safety layer, Vehicle-to-grid, Zero-shot transfer

1 Introduction

A charge point operator (CPO) buys energy for its chargers at time-varying prices, and vehicle-to-grid (V2G) discharging lets the same operator sell energy from parked electric vehicles (EVs) when prices are high. Two hard requirements limit this freedom. The aggregate power of the station must stay within the transformer limit, and each EV must hold its agreed energy when it leaves. Otherwise the operator overloads equipment or breaks the service promise to the driver.

Reinforcement learning (RL) has become a common approach to this control task, because prices, arrivals and departures are uncertain and hard to model in advance [1, 2]. Most RL controllers for EV charging encode the departure energy requirement as a weighted penalty in the reward [3–6]. Those that model a transformer limit usually penalise overloads in the same way [6, 7]. A penalty trades violations against profit, so a trained policy may still violate either requirement at test time [6,8]. Constrained formulations

bound the expected cumulative cost instead [9], but such a bound holds on average over episodes and not at every step [10].

A second limitation concerns station size, because a flat observation with one action per charger, as in the default interface of the EV2Gym simulator [6], ties a trained network to one number of chargers. Graph and attention policies remove this restriction [7, 11, 12], yet in these size-agnostic controllers the constraints still enter through reward terms, imitation data or heuristic filters.

Safety layers correct each action before execution and can enforce limits by construction [8, 13, 14]. For EV stations, existing layers cover per-step limits such as transformer capacity and charger current [15], or they replace unsafe actions [16, 17]. A per-step layer alone does not protect the departure energy requirement, because that requirement couples the present action with every later step of the stay. Per-EV latest-start bounds address this coupling in laxity-based scheduling [18] and in a recent hierarchical V2G scheduler [19]. Section 2.5 positions the combination of these elements against the closest works.

This paper studies size-agnostic safe control for V2G charging at a CPO station. A set policy treats each charger as a token and outputs one raw power action per token with shared weights. The main instantiation, the safe attention-based controller (SA-Attn), uses self-attention, and a Deep Sets policy serves as the alternative set architecture. A safety layer, an exact projection onto the feasible set, maps the raw actions to commanded powers that respect per-EV bounds and the transformer limit. The projection onto a box intersected with a two-sided sum constraint is classical [20, 21], and the rate rule of smoothed least-laxity-first (sLLF) scheduling has a closely related form [18]. Our feasibility argument extends the laxity argument of Chen et al. to the charging taper, V2G discharging and a two-sided coupling.

The paper makes three contributions.

1. We derive a safety layer whose per-EV lower bound includes the charging taper, and Theorem 1 states its result in two layers. The transformer limit holds at every step for every sequence of raw actions under simulator and grid-headroom conditions, and the admitted departure requirement is met by every EV whose stay contains no infeasible step.
2. We train two size-agnostic set policies, an attention policy and a Deep Sets policy, with the layer in the loop on stations with 10 and 20 chargers. Both are tested without retraining on up to 200 chargers, against a reference policy trained at 50 chargers.
3. We report a measured evaluation in EV2Gym of the transformer limit with and without the layer and of the conditional departure result at episode level. It also measures how often learned policies cause infeasible steps, the cost of the guarantee, simple rules with the same layer and a split-conformal departure bound.

The results do not show an advantage of the attention policy over the Deep Sets policy. A price-threshold rule with the safety layer had a lower mean gap than the attention policy at every station size of the test grid. Only the Deep Sets policy beat this rule, and only from 20 chargers upward. For the attention policy, the departure bound also came with more unserved EVs than a variant without it. None of the ingredients is new in isolation, so the claim of the paper rests on their combination, on the two-layer result and on its measured behaviour.

Section 2 reviews related work, Section 3 states the model as a partially observable Markov decision process (POMDP), and Section 4 presents the method. Sections 5 to 8 cover the setup, results, discussion and conclusion, and the supplementary material holds all proofs and further details.

2 Related Work

2.1 Reinforcement learning for EV charging and V2G

Deep RL for EV charging has moved from single-vehicle agents [22] to stations with photovoltaics, storage and V2G [5, 17], to coordinated stations [23] and to fleets under distribution-grid limits [24, 25]. Two reviews list safety, dependence on simulation and the lack of a standard benchmark among the open problems [1, 2]. EV2Gym models a CPO whose chargers sit under transformers with inflexible load

and solar generation [6]. Its V2G reward penalises transformer overload and low user satisfaction, and in its V2G benchmark penalty-based agents earned profit but left many users short of their requested energy. The Adaptive Charging Network (ACN) solves a convex program in receding horizon with hard infrastructure limits and best-effort energy delivery [26]. An economic model predictive control (MPC) inside EV2Gym enforces hard departure and transformer constraints [27].

2.2 Constraint handling and safe reinforcement learning

Weighted violation terms in the reward do not prevent violations at test time [6, 8]. The constrained Markov decision process (CMDP) bounds an expected cumulative cost instead [9], solved by Lagrangian [28] or trust-region [29] updates. Reviews note that such bounds hold for an expected cumulative cost rather than at each step [10] and classify Lagrangian methods as soft guarantees [30]. CMDP formulations have been applied to EV charging with voltage limits [31], transformer thermal stress [32, 33] and V2G aggregators [34]. A Lagrangian multi-agent method also encodes the recent history of each station with a transformer [35].

A second family corrects each action before execution, through shields [36], learned linear safety layers [13] or quadratic programs with a fallback controller [37]. EV examples include rule-based shields [16], action-replacement layers [17], grid-code shielding [38] and power-flow shields [39]. Such a filter is only as safe as its internal model, and the authors of the HetGPS preprint make no pointwise feasibility claim for its learned risk model with a linearised power-flow projection [12]. Members of the EV2Gym team have also combined a graph-enhanced twin delayed deep deterministic policy gradient (TD3) agent [40] with a safety layer [41]. According to the chapter abstract, the layer is intended to preserve charging requirements and enforce power limits.

A third family makes the policy output feasible by construction. Differentiable convex optimisation layers pass gradients through an exact solve [42, 43], and Chen et al. used such a projection inside energy-system policies [14]. Gauge maps give feasible points directly [44], and Tabas and Zhang used one to prove that a learned frequency controller stays in a controlled-invariant safe set [8]. For EV stations, a preprint by Wan et al. proves per-step feasibility of a quadratic-program projection that covers transformer and charger-current limits but not departure energy [15]. A preprint by Yan et al. derives per-EV bounds from remaining energy and time, and its formal result concerns first-order power-balance feasibility [19]. Projection also alters learning, because naive projection biases policy gradients [45] and an active projection has zero gradients [46]. For stochastic policies, a recent preprint shows that projection inside the environment is equivalent to projection as a policy layer [47].

2.3 Size-agnostic architectures

Deep Sets [48], self-attention [49] and the Set Transformer [50] process sets of any size with shared weights, and UPDeT applied this idea to multi-agent tasks of different sizes [51]. Size transfer is not automatic, since neural routing solvers trained on small graphs often fail on larger ones [52]. Early size-independent states in EV charging came from aggregation, and Sadeghianpourhamami et al. also reported the gap to a perfect-knowledge optimum under scaled arrival rates [3]. Aggregated representations, however, hide per-vehicle detail or move it into disaggregation rules [53, 54].

The preprint version of the heterogeneous graph policy (EV-GNN) coordinates up to 1000 charge points, with a separate model per scale and no constraint enforced by construction [11]. In contrast, the graph-enhanced decision transformer (GNN-DT), trained on 25 chargers, was tested without retraining on 5 to 100 chargers, and its performance degraded as size grew [7]. A HetGPS policy trained with 1,236 vehicles transfers without retraining to feeders with 2,239 and 3,218 vehicles [12]. Similarly, a graph actor for voltage control has been evaluated in simulation on a larger feeder than in training [55].

In these size-agnostic controllers, constraints enter through reward terms, imitation data or heuristic filters.

Table 1: Positioning against the closest works. "Penalty" means a reward term; "hard" means enforced by construction.

Work	Size-agnostic policy	Departure energy	Transformer limit	Guarantee	Zero-shot size test	Gap to optimum
EV2Gym agents [6]	no (flat)	penalty	penalty	none	no	optimum as baseline
EV-GNN, preprint version [11]	yes (graph)	not by construction	not by construction	none	separate model per scale	optimisation baseline per scale
GNN-DT [7]	yes (graph)	penalty	penalty	none	25 to 5–100 chargers	near-oracle reward (journal abstract)
HetGPS, preprint [12]	yes (graph attention)	reward and heuristic	not constrained	empirical (voltage filter)	1,236 to 3,218 vehicles	no
Yan et al., preprint [19]	no	hard (per-EV rule)	not modelled	first-order power balance	no	gap to MPC
Wan et al., preprint [15]	no	not modelled	hard (quadratic-program projection)	per-step feasibility	no	no
sLLF [18]	rule, any size	laxity rule	hard (common level)	feasibility theorems	not learned	no
Sadeghian-pourhamami et al. [3]	yes (aggregate state)	penalty	not modelled	none	scaled arrivals	yes
Orfanoudakis et al. [41] (abstract)	graph policy	safety layer (stated aim)	safety layer (stated aim)	not stated in abstract	not stated in abstract	not stated in abstract
This work	yes (attention and Deep Sets over charger tokens)	per-EV bound, conditional (Thm. 1b)	hard (common shift)	transformer: every step (A3, A6); departure: per EV, conditional	10/20 to 200 chargers	soft- and hard-constrained optima

2.4 Laxity-based scheduling and departure uncertainty

Chen et al. proved that a schedule is feasible exactly when every laxity, the slack between the time left and the time still needed to finish charging, stays non-negative [18]. Their sLLF rule clips each rate around a common level under the station power limit, and their bounds indicate that online feasibility generally requires extra power. A preprint shows that hard deadlines and a transformer limit can become jointly infeasible under congestion [56], and least-laxity-first (LLF) priorities perform well under cable and transformer limits [57, 58]. These methods assume that users declare departure time and energy at arrival. Plug-in durations are hard to predict [59], users can gain by under-reporting deadlines [60], and departure forecasts improve prioritisation [61]. A recent preprint trains a departure forecaster end-to-end with the charging policy but keeps unmet energy as a soft penalty and gives no coverage statement [62]. Conformal prediction gives distribution-free coverage under exchangeability [63, 64] and has tightened constraints in safe planning [65].

2.5 Positioning

As seen in Table 1, each ingredient of our design already exists in some form. Size-agnostic policies with zero-shot transfer exist for EV charging [7,12], provable action-correction layers exist in safe RL [8,15], and a closely related clip-around-a-common-level rule appears in sLLF [18]. Per-EV latest-start bounds appear in the scheduler of Yan et al. [19], and train-small, test-large evaluation with a gap to an optimum dates back at least to [3]. Three closely related recent works were available to us only as abstracts [23, 34, 41], and we describe them only at that level. To our knowledge, no prior work joins a size-agnostic learned policy, a per-step transformer guarantee and a multi-step departure argument in one controller.

3 Problem Formulation

3.1 Station, EVs and energy dynamics

A CPO runs a station with N chargers $\mathcal{N} = \{1, \dots, N\}$ behind one transformer, and each charger hosts at most one EV, whose quantities carry the index i of its charger. Time is divided into steps $t = 0, \dots, T-1$ of length Δt [h]. At every step the station observes the transformer rating $P_{\mathrm{tr}}(t)$, the inflexible load $P^{\mathrm{inf}}(t) \geq 0$ and the photovoltaic (PV) generation $P^{\mathrm{pv}}(t) \geq 0$ in kW. Energy is bought at the price $c^{\mathrm{ch}}(t)$ and sold at the price $c^{\mathrm{dis}}(t)$ [€/kWh], and both prices may be negative.

EV i arrives at step t_i^{arr} and is controlled until its departure step d_i, the last controlled step, so its departure energy is $E_i(d_i + 1)$, with $E_i(t)$ the battery energy at the start of step t. At step t the EV has $\tau_i(t) = d_i - t + 1$ remaining actions, and $\mathcal{N}(t)$ is the set of the $n(t)$ occupied chargers. Let $p_i(t)$ [kW] be the power commanded at charger i, positive for charging and negative for V2G discharging, and let $\tilde{p}_i(t)$ be the power reported by the simulator. With $p^+ = \max(p, 0)$ and $p^- = \max(-p, 0)$, the battery energy evolves as

$$E_i(t+1) = E_i(t) + \Delta t \, g_i\big(p_i(t)\big), \qquad g_i(p) = \eta_i^{\mathrm{c}} p^+ - p^- / \eta_i^{\mathrm{d}}, \tag{1}$$

with charge and discharge efficiencies $\eta_i^{\mathrm{c}}, \eta_i^{\mathrm{d}} \in (0, 1]$. Per-step physical limits restrict the power to

$$p_i^{\min}(E) \leq p_i(t) \leq \bar{P}_i(E), \qquad p_i^{\min}(E) = -\min\Big\{P_i^{\mathrm{dis}},\ \eta_i^{\mathrm{d}} \frac{(E - E_i^{\min})^+}{\Delta t}\Big\}, \qquad E = E_i(t), \tag{2}$$

where the maximum charging power $\bar{P}_i(E) \geq 0$ includes the charging taper and the capacity limit, $\bar{P}_i^{\mathrm{nom}} = \sup_E \bar{P}_i(E)$ and P_i^{dis} is the maximum discharging power. The theory holds for the general map g_i, but EV2Gym reports, bills and meters the battery-side energy change as the charger power [6]. Our experiments therefore use $\eta_i^{\mathrm{c}} = \eta_i^{\mathrm{d}} = 1$, a convention that removes the round-trip loss of V2G arbitrage (Section 7.2).

3.2 Departure requirement and transformer limit

Each EV requests a desired departure energy E_i^{des}, and a station-level request pipeline turns it into the admitted requirement E_i^{req}. The pipeline runs once per test replay before any controller, and every compared method and both optima receive the same admitted requirement. At arrival it caps the request to the energy that full-power charging can reach during the stay, about 10^{-6} kWh below that value. Admission control then solves a small feasibility linear program (LP) and lowers the new request if the connected EVs could otherwise not all be served under the forecast transformer limits. Requests already admitted are never reduced. Hence

$$E_i^{\mathrm{req}} \leq \min\big\{E_i^{\mathrm{des}},\ \text{the energy reachable at full power from } E_i^{\mathrm{arr}} \text{ during the stay}\big\}, \tag{3}$$

and Section S3 gives the details. The departure requirement is

$$E_i(d_i + 1) \ \geq \ E_i^{\mathrm{req}}. \tag{4}$$

Without the cap, some arrivals could not be served by any controller, and a guarantee about them would be void. For every method, shortfall is measured against the shared admitted requirement, and the energy removed by the pipeline is reported separately.

The net transformer power $P^{\mathrm{inf}}(t) - P^{\mathrm{pv}}(t) + \sum_i p_i(t)$ must stay within $[-P_{\mathrm{tr}}(t), P_{\mathrm{tr}}(t)]$, so the EV aggregate has the transformer limit

$$-C_{\mathrm{rev}}(t) \leq \sum_{i \in \mathcal{N}} p_i(t) \leq C(t), \qquad C(t) = P_{\mathrm{tr}}(t) - P^{\mathrm{inf}}(t) + P^{\mathrm{pv}}(t), \qquad C_{\mathrm{rev}}(t) = P_{\mathrm{tr}}(t) + P^{\mathrm{inf}}(t) - P^{\mathrm{pv}}(t). \tag{5}$$

The forward limit $C(t)$ bounds charging and the reverse limit $C_{\mathrm{rev}}(t)$ bounds V2G export. For the current step the safety layer uses the measured load and PV, whereas the policy observes only noisy forecasts of future values.

3.3 Online control problem and optima

At each step the CPO chooses $p(t) \in \mathbb{R}^N$ using only the information available at step t. The profit of a controller π in an episode, in €, is

$$J^{\pi} = \sum_{t=0}^{T-1} \Delta t \sum_{i \in \mathcal{N}(t)} \big(c^{\mathrm{dis}}(t)\, \tilde{p}_i^{-}(t) - c^{\mathrm{ch}}(t)\, \tilde{p}_i^{+}(t) \big), \tag{6}$$

and the controller should maximise its expectation subject to Equations 1–2, Equation 4 for every EV and Equation 5 at every step. With the weights $w_E = w_P = 10\,€/\mathrm{kWh}$ for every method, the penalised profit is

$$J^{\mathrm{pen}}(\pi) = J^{\pi} - w_E \sum_i \Delta E_i^{\mathrm{sh}} - w_P\, \Delta t \sum_t \Big[\Big(\sum_i \tilde{p}_i(t) - C(t) \Big)^{+} + \Big(-C_{\mathrm{rev}}(t) - \sum_i \tilde{p}_i(t) \Big)^{+} \Big], \tag{7}$$

where $\Delta E_i^{\mathrm{sh}} = (E_i^{\mathrm{req}} - E_i(d_i + 1))^{+}$ is the departure shortfall.

Two perfect-information LPs, solved with HiGHS [66], know the whole episode and serve as references. Both model EV2Gym's taper with outer linear rows, exact except on a one-step transition band and a 1% margin (Lemma S8), and both allow for EV2Gym's unbilled 0.01 kWh rounding gain. The soft-penalty optimum $J^{\mathrm{pen,LP}}$ maximises Equation 7 with slack variables for shortfall and overload. For every simulated trajectory, with shortfall measured against the shared admitted requirement, $J^{\mathrm{pen}}(\pi) \le J^{\mathrm{pen,LP}}$ (Proposition S3), including trajectories with shortfall or overload. The main metric is therefore well defined for every compared method, as the absolute gap in € per charger and episode,

$$\mathrm{gap}^{\mathrm{pen}}(\pi) = \frac{J^{\mathrm{pen,LP}} - J^{\mathrm{pen}}(\pi)}{N} \ge 0. \tag{8}$$

A relative gap is not used per episode, because $J^{\mathrm{pen,LP}}$ is often close to zero or negative, and the supplementary material reports relative gaps only as ratios of sums. The hard optimum $J^{\mathrm{LP,ceil}}$ enforces the departure requirements and the transformer limit as constraints, after a first stage that minimises the total shortfall. It bounds the profit of every controller that respects the transformer limit and leaves no more shortfall (Proposition S2), and it serves the analysis of the cost of the guarantee.

3.4 Partially observable Markov decision process

The online problem is a POMDP, because future arrivals, prices beyond a short window and future loads are not observed. We act on the current observation with a feed-forward actor and critic. The policy outputs one raw action $a_i(t) \in \mathbb{R}$ per charger, scaled to kW as $x_i(t) = P^{\mathrm{ref}} a_i(t)$ with $P^{\mathrm{ref}} = 11$ kW, and the safety layer of Section 4.3 maps it to the commanded power without any squashing function. Its reward is the realised profit of the step, with no penalty for the departure requirement or the transformer limit, because the safety layer enforces both. For training, the reward is divided by $N r^{\mathrm{ref}}$ with $r^{\mathrm{ref}} = c^{\mathrm{ref}} P^{\mathrm{ref}} \Delta t$ and $c^{\mathrm{ref}} = 0.25\,€/\mathrm{kWh}$, so that returns do not grow with N, and all reported results use the undivided profit. Arrivals, prices, loads and PV evolve independently of the actions, every EV departs before the episode ends, and the discount factor $\gamma = 0.99$ gives an effective horizon of about one day at $\Delta t = 15$ min.

4 Proposed Method

As seen in Figure 1, SA-Attn has three parts that act in sequence at every step. An attention policy maps a set of charger tokens to one raw action per charger, and a safety layer turns the raw actions into commanded powers that respect per-EV bounds and the transformer limit. During training, a policy-gradient method updates the policy from the realised profit with the safety layer treated as part of the environment, and the same layer acts unchanged during deployment.

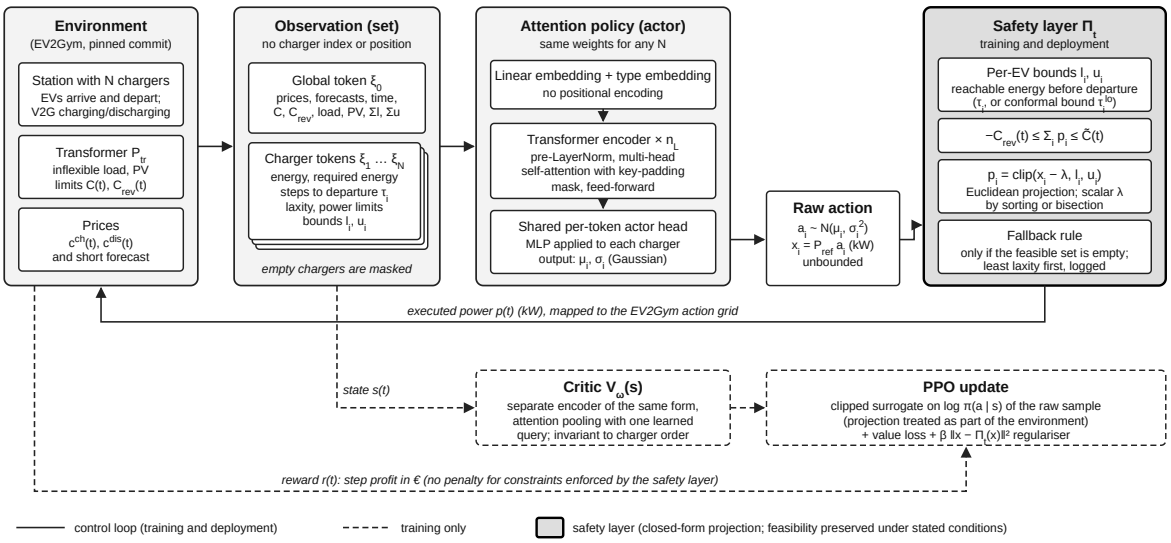


Figure 1: Architecture and control loop of SA-Attn. Solid arrows form the control loop used in training and deployment; dashed arrows are used in training only. The safety layer (shaded) is an exact projection that keeps the transformer limit and keeps each EV's requirement reachable while the EV receives its lower bound (Theorem 1).

4.1 Token representation and set policies

The observation is a set of N charger tokens $\xi_i(t) \in \mathbb{R}^{12}$ and one global token $\xi_0(t) \in \mathbb{R}^{38}$, and no feature encodes the index or the position of a charger. A charger token holds occupancy, energy and capacity data, the remaining required energy and steps, the laxity of Section 4.2, the power limits and the bounds l_i and u_i. The global token holds prices for the next $H_f = 8$ steps, time features, the current limits, load and PV forecasts and four station ratios. All features are normalised so that they do not grow with N, and Table S3 lists them.

Each token is embedded linearly with one of two learned type embeddings, $e_i = W_c\xi_i + b_c + v_c$ for chargers and $e_0 = W_g\xi_0 + b_g + v_g$ for the global token. The sequence $[e_0; e_1; \ldots; e_N]$ passes through $n_L = 3$ pre-LayerNorm transformer encoder layers [49, 67, 68] of width $d_m = 128$ with $n_H = 4$ heads and no positional encoding. A key-padding mask removes empty chargers as attention keys, and the global key is never masked. Then a shared actor head maps each charger output h_i to the mean $\mu_{\theta,i}$ and standard deviation $\sigma_{\theta,i}$ of a Gaussian raw action, where θ denotes the actor parameters. The policy is $\pi_\theta(a \mid s) = \prod_{i\in\mathcal{N}(t)} \mathrm{Normal}\big(a_i; \mu_{\theta,i}(s), \sigma^2_{\theta,i}(s)\big)$. A separate critic encoder with parameters ω pools all unmasked token outputs with one learned attention query [50] into the value $V_\omega(s)$.

The construction is permutation-equivariant, masked tokens do not influence the outputs of unmasked tokens, and the number of parameters does not depend on N (Lemma S1). Whatever the number of chargers, the actor has 610,370 parameters and the critic 676,481. Weights trained at $N \in \{10, 20\}$ can therefore run at $N = 200$, but performance there is not guaranteed, because attention over 200 tokens averages over more keys than seen in training. The Deep Sets alternative replaces the encoder by a shared per-token network with the masked mean of all token embeddings and the global token as context, so no token attends to another [48]. Its width is chosen to match the parameter count of SA-Attn within 10%, and every other part of the method is unchanged.

4.2 Per-EV bounds from reachable energy

For each connected EV, the safety layer needs the least power that keeps its departure requirement reachable, and we drop the index i where no confusion arises. Full-power charging for one step maps the energy E to $\varphi(E) = E + \Delta t\, \eta^{\mathrm{c}}\, \bar{P}(E)$. Assumption A1 of Section 4.4 asks that φ is continuous and non-decreasing, so that starting a step with more energy never leaves less energy after full-power charging. The backward-reachable thresholds, the least energies from which the requirement is still reachable at

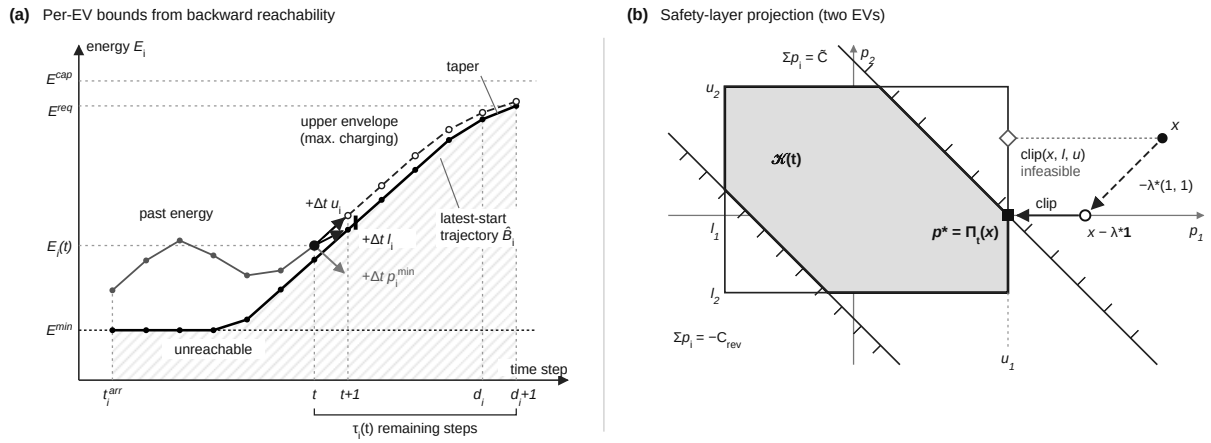


Figure 2: Concepts of the safety layer. (a) Energy of one EV during its stay, with the latest-start trajectory $\hat{B}_i$ and the per-step options $\Delta t\, u_i$, $\Delta t\, l_i$ and $\Delta t\, p_i^{\min}$. (b) Projection for two EVs onto the box $[l, u]$ cut by the slab $-C_{\mathrm{rev}} \le p_1 + p_2 \le \tilde{C}$.

full power, are

$$B(0) = E^{\mathrm{req}}, \qquad B(k) = \min\big\{E \in [E^{\min}, E^{\mathrm{cap}}] :\ \varphi(E) \ge B(k-1)\big\}, \quad k \ge 1. \tag{9}$$

Under A1, an EV is individually feasible at step t, in the sense that some admissible control meets Equation 4, exactly when $E(t) \ge B(\tau(t))$ (Lemma S2). Without A1 this characterisation fails, because charging less now could leave more reachable energy later. For EV2Gym's exponential taper, Lemma S6 proves A1 for every step length.

For a connected EV with energy $E = E_i(t)$ and $\tau = \tau_i(t)$ remaining actions, the bounds of the safety layer are

$$u_i(t) = \bar{P}(E), \qquad l_i(t) = \begin{cases} \min\Big\{u_i(t),\ \max\Big\{p^{\min}(E),\ g^{-1}\Big(\dfrac{\hat{B}(\tau-1) - E}{\Delta t}\Big)\Big\}\Big\}, & E \ge \hat{B}(\tau), \\ \bar{P}(E), & E < \hat{B}(\tau), \end{cases} \tag{10}$$

where $\hat{B}$ is a stored threshold table and the outer minimum is a numerical guard. In the first case, l_i is the smallest admissible power after which $\tau - 1$ full-power steps still reach E^{req}, and an individually infeasible EV is charged at maximum power. A negative lower bound allows V2G discharging while slack remains, and as seen in Figure 2(a), the bound keeps the energy on or above the latest-start trajectory.

The table $\hat{B}_i(k)$, $k = 0, \dots, \tau_i(t_i^{\mathrm{arr}})$, is computed once at arrival by bisection on φ_i, so the bounds cost $O(1)$ per EV and step. Each stored table is certified in the sense that

$$\hat{B}(0) \ge E^{\mathrm{req}}, \qquad \varphi\big(\hat{B}(k)\big) \ge \hat{B}(k-1), \qquad \hat{B}(k) \le \max\{\hat{B}(k-1), E^{\min}\}, \qquad \hat{B}(k) \in [E^{\min}, E^{\mathrm{cap}}], \tag{11}$$

and the proofs use only these properties. The laxity $\chi_i(t) = \big(E_i(t) - \hat{B}_i(\tau_i(t))\big) / \big(\eta_i^{\mathrm{c}} \bar{P}_i^{\mathrm{nom}} \Delta t\big)$, measured in steps, is non-negative exactly when the EV is individually feasible.

4.3 Projection

Let $x \in \mathbb{R}^N$ be the raw action in kW, with $x_i = l_i = u_i = 0$ for empty chargers. The safety layer uses the tightened forward limit $\tilde{C}(t) = C(t) - n(t)\varepsilon_q$, where the margin $\varepsilon_q = 0.008$ kW per charger absorbs the actuation rounding of the simulator (Section S2.2). Its feasible set is $\mathcal{K}(t) = \{p \in \mathbb{R}^N :\ l \le p \le u,\ -C_{\mathrm{rev}}(t) \le \mathbf{1}^\top p \le \tilde{C}(t)\}$, and the safety layer is the Euclidean projection

$$\Pi_t(x) = \arg\min_{p \in \mathcal{K}(t)} \tfrac{1}{2}\|p - x\|_2^2. \tag{12}$$

Proposition 1 (Projection)**.** *(i) The set $\mathcal{K}(t)$ is non-empty if and only if $l \le u$, $\sum_i l_i \le \tilde{C}$, $\sum_i u_i \ge -C_{\mathrm{rev}}$ and $-C_{\mathrm{rev}} \le \tilde{C}$.*

(ii) If $\mathcal{K}(t) \neq \emptyset$, the unique minimiser of Equation 12 is $p_i^* = \mathrm{clip}(x_i - \lambda^*, l_i, u_i)$. The common shift λ^* is zero if the clipped point already satisfies the aggregate limit, and otherwise it is the root of $\lambda \mapsto \sum_i \mathrm{clip}(x_i - \lambda, l_i, u_i)$ at the violated limit.

The proof verifies the Karush–Kuhn–Tucker (KKT) conditions, and the supplementary material also gives the Jacobian (Proposition S1). As seen in Figure 2(b), the projection shifts all free coordinates by the same amount and clips them until the aggregate limit holds. One sort of the $2N$ breakpoints $\{x_i - u_i\} \cup \{x_i - l_i\}$ and one pass over them give λ^*, so the layer costs $O(N \log N)$ per step. A post-check with a bisection fallback keeps the aggregate limit under floating-point rounding, and Algorithm S1 lists one complete step.

The algebra is not new: Equation 12 is the continuous quadratic knapsack problem [20, 21], and the sLLF rate rule uses a closely related form, a common level scaled per EV by its peak rate [18]. Our use differs in its input, a learned per-charger action of a size-agnostic policy, and in the lower bound l_i from backward reachability with taper and V2G sign range. Moreover, the aggregate constraint is two-sided, and the layer acts during training.

4.4 Feasibility preservation

Intuitively, the safety layer never lets an EV fall below the least energy from which its requirement is still reachable at full power, and the common shift keeps the station inside the transformer band. Only the first property can fail, namely when the transformer cannot give every EV its lower bound at once. We call a step feasible when $\mathcal{K}(t)$ is non-empty, and on an infeasible step a fallback rule replaces the projection. The conditions of the result are collected below, and only the first five are assumptions. Technically, the induction behind it extends the laxity argument of Chen et al. [18] to the taper, V2G discharging, a two-sided coupling and arbitrary raw actions.

Conditions of Theorem 1

Label	Condition	Status in this study
A1	φ continuous and non-decreasing	proved for EV2Gym's taper (Lemma S6)
A2	departure step d_i and requirement E_i^{req} known at arrival	relaxed by the conformal bound (Section 4.6)
A3	one-step model: $E_i(t+1) \geq E_i(t) + \Delta t\, g_i(p_i)$ and $p_i \leq \tilde{p}_i \leq p_i + \varepsilon_q$ for commands within Equation 2	proved for EV2Gym's actuation (Lemma S7); may fail on hardware
A4	feasible arrival: $E_i^{\mathrm{arr}} \geq \hat{B}_i(\tau_i(t_i^{\mathrm{arr}}))$	enforced by the request pipeline
A6	grid headroom: $C(t) \geq N\varepsilon_q$ and $C_{\mathrm{rev}}(t) \geq 0$	exogenous; checked through the fallback counts
A5	feasible step: $\sum_i l_i \leq \tilde{C}$, $\sum_i u_i \geq -C_{\mathrm{rev}}$, $\tilde{C} \geq -C_{\mathrm{rev}}$	a closed-loop event, not assumed by (a) and (b); measured

A6 uses the number of chargers N instead of $n(t)$, so that it does not depend on the policy, and it fails in our scenarios with a probability below 2×10^{-9} per episode (Section S2.3).

Theorem 1 (Feasibility preservation)**.** *Let the commanded powers be $p(t) = \Pi_t(x(t))$ on feasible steps and the output of the fallback rule otherwise, for an arbitrary sequence of finite raw actions, during training or deployment.*

(a) Under A3 and A6, the realised powers satisfy the transformer limit $-C_{\mathrm{rev}}(t) \leq \sum_i \tilde{p}_i(t) \leq C(t)$, and the commanded powers satisfy the physical limits of Equation 2, at every step.

(b) Under A1–A4, every EV i with $p_i(t) \geq l_i(t)$ throughout its stay meets $E_i(d_i{+}1) \geq E_i^{\mathrm{req}}$, in particular every EV whose stay contains no infeasible step.

(c) If, in addition, every step is feasible, every EV meets its departure requirement.

Proof sketch. On feasible steps, Proposition 1(i) gives $p(t) \in \mathcal{K}(t)$, and A3 transfers the tightened band to the realised powers. Under A6, the fallback rule can only meet the case of excess required power described below, and its output keeps the commanded sum at $\tilde{C}(t)$. For (b), a one-step lemma (Lemma S3) carries the invariant $E_i(t) \geq \hat{B}_i(\tau_i(t))$ from arrival, where A4 gives it, through every step with $p_i \geq l_i$, and at

the last step it gives $E_i(d_i+1) \geq \hat{B}_i(0) \geq E_i^{\mathrm{req}}$. Part (c) follows from (b), and Section S1.5 gives the full proof. □

The transformer part depends neither on the policy nor on A1 or A5, whereas the departure part is a per-EV statement. A5 can fail even when a joint plan exists. Two EVs that each need 10 kWh within two 1-h steps under a 10 kW limit can share the capacity equally. A policy that idles in the first step, however, leaves $\sum_i l_i = 20$ kW in the second step. We therefore measure how often EVs meet infeasible steps.

4.5 Fallback rule

On an infeasible step, the fallback rule replaces the projection with a deterministic least-laxity-first allocation. In the relevant case, the required power $\sum_i l_i$ exceeds $\tilde{C}$. Every EV then starts at its physical lower limit $p_i^{\min}$, and the remaining budget up to $\tilde{C}$ goes to the EVs in increasing order of laxity, each up to l_i. This output keeps the transformer limit, and every feasible EV that still receives l_i stays feasible (Lemma S4). Three further cases cover contradictory limits, excess PV export and a negative budget, and A6 excludes all three (Section S1.6). For the EVs that receive less than l_i, the total departure shortfall of an episode is bounded a priori by the accumulated deficit of these steps (Lemma S5). Admission control in the request pipeline makes infeasible steps rarer, but it relies on forecasts and does not enter Theorem 1.

4.6 Conformal lower bound on the departure step

To relax A2 for the departure step, we use split conformal prediction (CP) [63, 64, 69, 70] in a one-sided, quantile-regression form [71]. A gradient-boosting regressor $\hat{q}$ estimates a lower quantile of d_i from the arrival step, the weekday, the arrival energy, the requested energy and the EV model. Calibration draws one session uniformly from each of n_{cal} independently seeded episodes and uses the scores $e_j = \hat{q}(\mathbf{w}_j) - d_j$. With $\hat{Q}_{1-\alpha}$ the $\lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil$-th smallest score, the bound $\hat{d}_i^{\mathrm{lo}} = \max\{t_i^{\mathrm{arr}}, \lceil \hat{q}(\mathbf{w}_i) - \hat{Q}_{1-\alpha} \rceil\}$ replaces d_i in the bounds, the request pipeline and the features, and after $\hat{d}_i^{\mathrm{lo}}$ the layer uses one remaining step.

Proposition 2 (Conformal departure guarantee)**.** *Let the calibration sessions and the test session be exchangeable. Then* $\mathbb{P}(d_i \geq \hat{d}_i^{\mathrm{lo}}) \geq 1-\alpha$*, and under A1, A3 and A4 with* $\hat{d}_i^{\mathrm{lo}}$ *in place of* d_i*,* $\mathbb{P}\big(E_i(d_i+1) \geq E_i^{\mathrm{req}}\big) \geq 1 - \alpha - \mathbb{P}\big(\exists\, t \leq d_i :\ p_i(t) < l_i(t)\big)$*.*

Here E_i^{req} is the requirement re-certified with $\hat{d}_i^{\mathrm{lo}}$, and the probability is over the calibration data and a uniformly chosen EV of a test episode, so the guarantee is marginal. Calibration needs data from each test configuration, so the conformal variant is not zero-shot.

4.7 Training algorithm

The policy is trained with proximal policy optimization (PPO) [72] and generalised advantage estimation [73], with the safety layer, the fallback rule and the actuation treated as part of the environment. PPO therefore uses the likelihood ratio of the raw Gaussian sample. The vanilla policy-gradient estimator, equal to the PPO surrogate gradient at the old parameters, is then unbiased for the return of the projected actions. For stochastic policies, projection inside the environment and inside the policy are equivalent in this sense [47], and the choice avoids the zero gradients of an active projection [46]. Tokens with $u_i - l_i \leq 10^{-9}$ kW cannot influence any commanded power and are dropped from the log-probability and the entropy. Fallback steps are masked out of the actor loss but kept for the critic. Many raw actions map to the same projected point, so on feasible steps a regulariser with weight $\beta = 0.01$ keeps the policy mean close to $\mathcal{K}(t)$,

$$\mathcal{L}_{\mathrm{reg}}(\theta) = \beta\, \mathbb{E}_s\Big[\frac{1}{n(t)} \sum_{i \in \mathcal{N}(t)} \Big(\frac{\bar{x}_i - \Pi_t(\bar{x})_i}{P^{\mathrm{ref}}}\Big)^2\Big], \qquad \bar{x} = P^{\mathrm{ref}} \mu_\theta(s). \tag{13}$$

By the Moreau identity, the gradient of $\frac{1}{2}\mathrm{dist}^2(x, \mathcal{K})$ is $x - \Pi_t(x)$, so $\Pi_t(\bar{x})$ is evaluated under stop-gradient [14, 47]. Each rollout runs eight environments with $N = 10$ and eight with $N = 20$, and Algorithm S2 gives the full training loop.

5 Experimental Setup

5.1 Simulator and scenarios

All experiments use EV2Gym [6] at commit `55ef368c` without changes to its source. A wrapper computes the observations, applies the shared admitted requirements, the safety layer, the fallback rule and the actuation, and logs all metrics without reading future arrivals. Each station has N chargers behind one transformer, and EVs are drawn from the eight V2G-capable EV models of the simulator with one constant efficiency per model (0.90, and 0.93 for one model). The desired energy is the full battery, prices are the Dutch day-ahead prices shipped with EV2Gym, and the discharge price equals the charging price. Table S2 lists all scenario parameters.

The grid quantities and the arrival rate scale linearly with N. In particular, the transformer rating is $P_{\mathrm{tr}} = \rho_{\mathrm{tr}} N$, with a tight level of $\rho_{\mathrm{tr}} = 4$ kW per charger as in EV2Gym's 25-charger V2G configuration and a moderate level of 7 kW. Inflexible load and PV are scaled with multipliers of mean $0.4 P_{\mathrm{tr}}$, and forecasts in the observation carry Gaussian noise with a standard deviation of 10%. The workplace, public and private arrival profiles of EV2Gym start at hours that let overnight sessions fit into the 28-h horizon of $T = 112$ steps of 15 min.

Training mixes $N \in \{10, 20\}$, the three profiles and the two ratings within every batch. The test grid covers $N \in \{5, 10, 20, 50, 100, 200\}$, both ratings and all profiles, plus a stress level of $\rho_{\mathrm{tr}} = 3$ kW at $N \in \{20, 200\}$ and EV2Gym's original 25-charger configuration, 43 configurations in total. Zero-shot therefore refers to the station size and to the stress level, whereas the profiles are seen in training. Each configuration has 100 fixed test episodes for $N \leq 50$ and 50 for $N \geq 100$, shared by all methods, so each policy runs 3,550 test episodes. The stress level is the only sensitivity study that was run.

5.2 Compared methods

Every method and every evaluation pass receives the same admitted requirements, and shortfall is always measured against them. Methods without the layer execute only the physical clip $\mathrm{clip}(x_i, p_i^{\min}, u_i)$ without l_i or the transformer coupling.

Among the learned methods, SA-Attn and Deep Sets+safety layer are the two size-agnostic set policies with the safety layer. Attention+penalty is the penalty-based twin of SA-Attn: it learns from the profit minus a weight w times the shortfall and overload energy, without the safety layer. Its constrained counterpart, Attention+PPO-Lagrangian, replaces the fixed weight by Lagrange multipliers with a cost limit of zero [9, 28]. With $l_i = p_i^{\min}$, Attention+safety layer without departure bound reduces the layer to a per-step transformer projection as in [15] and learns the departure requirement from the penalty reward. The penalty-trained checkpoints also run with the safety layer at deployment only, and SA-Attn and Attention+penalty also run without admission control, keeping the arrival-capped request. A multilayer perceptron (MLP) over tokens zero-padded to 20 slots, with and without the layer, is a fixed-size reference evaluated only at $N \leq 20$. As an in-distribution reference, SA-Attn is also trained at $N = 50$ and evaluated at $N \in \{50, 100, 200\}$.

Six rules complete the comparison, each with and without the safety layer: as fast as possible (AFAP), as late as possible (ALAP), round robin, LLF, sLLF [18] and a price-threshold rule. Our HiGHS re-implementation of the economic MPC of EV2Gym [27], eMPC-HiGHS, runs at every N with a 10-step horizon and knows the exact prices and arrivals inside it. The conformal variant SA-Attn-CP is trained once at $\alpha = 0.1$ and evaluated with bounds recalibrated for every $\alpha \in \{0.01, 0.05, 0.1, 0.2\}$, with an uncalibrated point prediction as a further pass. EV-GNN [11] was planned as an external baseline but was not run, and GNN-DT [7] was not ported, because it trains offline on datasets that an optimiser generates for each scenario. The planned ablations, penalty-weight sensitivities, further MPC variants and other sensitivity studies were not run either.

5.3 Metrics

The main metric is the absolute gap of Equation 8, valid for every method. Penalised profits use energy not served against the shared admitted requirement for every method and pass, and profit, energy not

served and transformer overload energy are reported separately. For the safety layer we report the share of EVs whose stay contains no infeasible step (E4), the infeasible steps per fallback case and the measured shortfall against the deficit bound of Lemma S5. Because the logs carry no per-EV flag for the hypothesis of Theorem 1(b), its departure part is checked at episode level.

The cost of the guarantee is the difference between the penalty-based twin and the policy with the safety layer on the same episodes, in J^{pen} as the primary and in profit as the secondary measure. It is a ratio of sums over matched episodes in € per admitted EV and in percent of $|J^{\mathrm{LP,ceil}}|$. A positive value means that the penalty-based twin scored higher. The transfer gap of SA-Attn to the reference trained at $N = 50$ separates transfer loss from method loss.

5.4 Tuning, statistics and compute

Each of the seven tuned learned methods received 12 configurations from a grid over learning rate, minibatch size and entropy coefficient. Penalty rewards add the weight $w \in \{1, 10, 100\}$ to the grid, and the Lagrangian method adds the multiplier step size. Every configuration was trained on the training mixture for 0.5×10^6 steps and evaluated on 24 validation episodes disjoint from the test replays, and selection used the validation J^{pen} at $w = 10$ for every method. The selected configuration was retrained with two seeds for 3×10^6 steps, and Table S4 lists the hyperparameters.

With two seeds, between-seed variability cannot be tested, so the analysis compares seed-averaged policies on the shared test replays. Tables report the mean of the seed-averaged policy with the range over the two seeds. Paired Wilcoxon signed-rank tests [74] on replay-level differences compare SA-Attn with each other method per station size. Holm's correction runs over all comparisons of one results table, up to 351 per family, and the sign of the rank-biserial correlation gives the direction. Confidence intervals of the mean come from a bootstrap that resamples test replays within each configuration, and the interquartile mean (IQM) of the gap is reported at the training sizes [75]. The Wilcoxon test concerns the median shift, whereas the interval concerns the mean, and a few episodes with large penalties can move the mean alone. Where the two disagree, we report both and call the comparison inconclusive. Results over few seeds can mislead [76], so the seed count is a limitation (Section S4.4).

The implementation uses PyTorch [77], a single-file PPO in the style of CleanRL [78] and the Adam optimiser [79], with the safety layer in double precision. All experiments ran on an AMD Ryzen 7 7700 with an NVIDIA GeForce RTX 5070 Ti (16 GB) under WSL2 Ubuntu 24.04 with Python 3.13.15 and PyTorch 2.14.1. Preparation took 1.6 h, tuning 7.9 h, training 10.2 h and evaluation 2.5 h, for 153,100 episodes in total (Section S4.5). Code, configurations and analysis scripts will be released with the paper [TODO: repository link].

6 Results

[TODO: The authors check every reported value against the analysis outputs and revise the wording of this section].

6.1 Main comparison

Table 2 compares all methods at the training sizes $N \in \{10, 20\}$, pooled over profiles and ratings. SA-Attn had a gap of 1.290 and 0.973 € per charger at $N = 10$ and $N = 20$, against 2.204 and 1.569 for Attention+penalty and 1.792 and 1.258 for Attention+PPO-Lagrangian. The differences to Attention+penalty were -0.915 $[-1.254, -0.601]$ and -0.595 $[-0.797, -0.412]$, with Holm-adjusted Wilcoxon p-values of 1.3×10^{-5} and 0.0014 and rank-biserial correlations of -0.24 and -0.19. Their IQMs were close, 0.712 against 0.751 and 0.719 against 0.748, so the mean advantage of SA-Attn came mostly from a minority of episodes with large penalties. Deep Sets+safety layer had a lower gap than SA-Attn, 0.984 and 0.722, with differences of 0.306 $[0.218, 0.397]$ and 0.251 $[0.220, 0.286]$ (Holm $p < 10^{-65}$). Attention+safety layer without departure bound reached 0.776 at both sizes, below SA-Attn by 0.514 $[0.372, 0.679]$ and 0.198 $[0.145, 0.259]$ (Holm $p < 10^{-45}$).

At $N = 10$, every rule and eMPC-HiGHS with the safety layer had a lower mean gap than SA-Attn, from

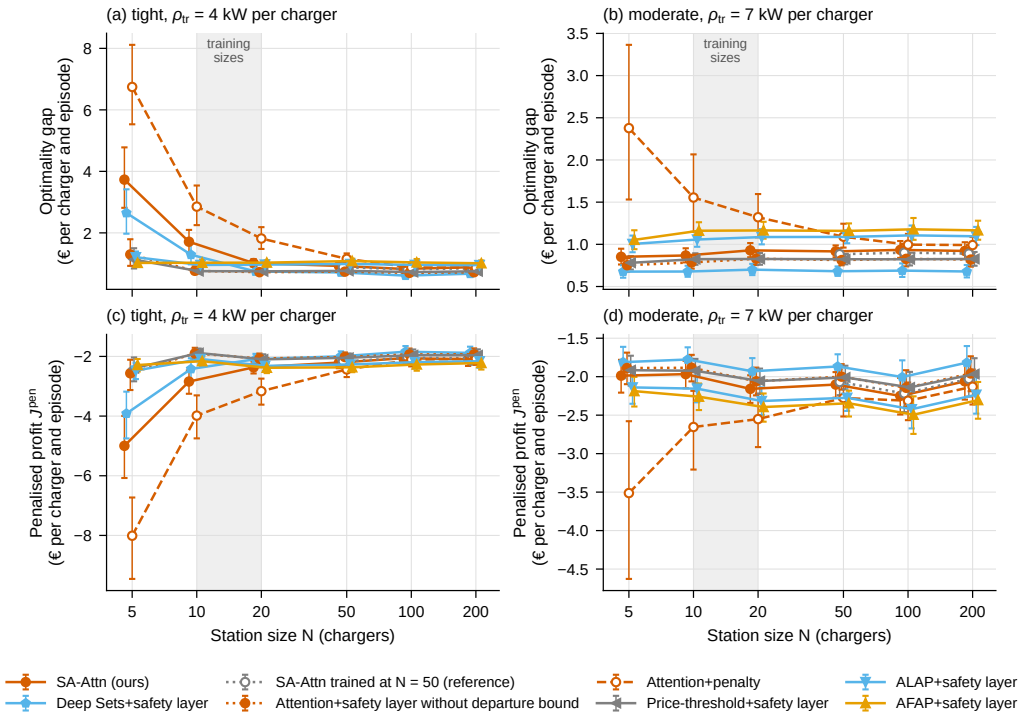


Figure 3: Zero-shot transfer across station sizes: gap and penalised profit per charger and episode versus N for the learned methods, the reference trained at $N = 50$ and three rules with the safety layer.

0.790 for the price-threshold rule to 1.145 for eMPC-HiGHS. For the price-threshold rule and eMPC-HiGHS, the differences of 0.500 $[0.341, 0.680]$ and 0.144 $[0.072, 0.219]$ agreed with the Wilcoxon tests (Holm $p = 1.1 \times 10^{-9}$ and 5.7×10^{-59}). With the other rules, the intervals excluded zero, but rank-biserial correlations of -0.39 to -0.47 showed SA-Attn lower on most replays, so these comparisons are inconclusive. MLP+safety layer also had a lower mean gap at $N = 10$, 1.025 (Holm $p = 0.014$).

At $N = 20$, SA-Attn had a lower gap than LLF, sLLF, AFAP and round robin with the layer, 0.973 against 1.099 (Holm $p < 10^{-24}$). Price-threshold+safety layer and eMPC-HiGHS+safety layer stayed lower, at 0.788 and 0.846, with differences of 0.185 $[0.137, 0.238]$ and 0.127 $[0.098, 0.161]$ (Holm $p \leq 9.8 \times 10^{-12}$). Without the safety layer, the price-threshold rule had gaps of 156.370 and 159.518 and eMPC-HiGHS 108.138 and 103.765, mostly from energy not served.

6.2 Zero-shot transfer across station sizes

Figure 3 and Table S5 show the gap against N for weights trained at $N \in \{10, 20\}$. At $N = 50$, 100 and 200, SA-Attn had gaps of 0.921, 0.881 and 0.902, and the same policy trained at $N = 50$ had 0.884, 0.857 and 0.869. The transfer gaps of SA-Attn to this reference were 0.037 $[0.026, 0.050]$, 0.023 $[0.012, 0.033]$ and 0.033 $[0.022, 0.046]$. Deep Sets+safety layer had gaps of 0.698, 0.649 and 0.675 at these sizes, below the reference by 0.186 to 0.208. At $N = 5$, outside the training sizes, the gap of SA-Attn rose to 2.292 against 1.663 for Deep Sets+safety layer and 0.956 for Price-threshold+safety layer.

SA-Attn had a higher mean gap than Price-threshold+safety layer at every size of the test grid. The differences ranged from 0.107 $[0.065, 0.151]$ at $N = 100$ to 1.335 $[0.919, 1.840]$ at $N = 5$, with Holm $p \leq 3.1 \times 10^{-4}$. Deep Sets+safety layer had a lower mean gap than this rule only at $N \geq 20$, with 0.649 to 0.722 against 0.774 to 0.793, and a higher one at $N = 5$ and $N = 10$. No test compares these two methods directly. Attention+penalty had a gap of 0.929 at $N = 200$, a difference to SA-Attn of -0.027 $[-0.060, 0.016]$. Its interval includes zero, whereas the Wilcoxon test favoured SA-Attn (Holm $p = 2.0 \times 10^{-5}$, rank-biserial -0.35), so this comparison is inconclusive.

At the stress level $\rho_{tr} = 3$ kW, SA-Attn had gaps of 2.979 at $N = 20$ and 1.586 at $N = 200$ (Table S15). Deep Sets+safety layer had 2.400 and 1.423 there, and the variant without departure bound 0.983 and 0.829.

Table 2: Main comparison at the training station sizes N ∈ {10, 20} (three arrival profiles and both tightness levels pooled; per episode). J^{pen}: penalised profit with 10 €/kWh for energy not served against the shared requirement and for transformer overload energy.

Method	Optimality gap (€ per charger and episode)	Optimality gap, IQM [95 % CI]	J^{pen} (€/episode)	Profit (€/episode)	Energy not served (kWh)	Overload (kWh)
N = 10						
SA-Attn (ours)	1.290 (1.255–1.325)	0.712 [0.664, 0.769]	−24.05	−19.42	0.46	0
Attention+penalty	2.204 (2.172–2.237)	0.751 [0.697, 0.807]	−33.19	−18.96	0.68	0.74
Attention+ PPO-Lagrangian	1.792 (1.330–2.255)	1.009 [0.939, 1.085]	−29.07	−21.89	2.7e-06	0.72
Attention+safety layer without departure bound	0.776 (0.733–0.819)	0.557 [0.518, 0.597]	−18.91	−18.20	0.07	0
Deep Sets+safety layer	0.984 (0.940–1.028)	0.535 [0.499, 0.574]	−20.99	−17.71	0.33	0
MLP+safety layer	1.025 (1.012–1.037)	0.679 [0.631, 0.729]	−21.40	−19.33	0.21	0
MLP+penalty	5.547 (2.529–8.564)	1.182 [1.082, 1.288]	−66.61	−21.79	3.73	0.75
LLF+safety layer	1.095†	0.890 [0.836, 0.951]	−22.10†	−22.10†	0†	0†
LLF	1.095†	0.890 [0.830, 0.950]	−22.10†	−22.10†	0†	0†
sLLF+safety layer	1.095†	0.890 [0.830, 0.948]	−22.10†	−22.10†	0†	0†
sLLF	1.095†	0.890 [0.829, 0.948]	−22.10†	−22.10†	0†	0†
Price-threshold+ safety layer	0.790†	0.622 [0.580, 0.668]	−19.04†	−18.85†	0.02†	0†
Price-threshold	156.370†	147.376 [139.977, 155.275]	−1574.85†	19.46†	153.20†	6.23†
AFAP+safety layer	1.095†	0.890 [0.833, 0.950]	−22.10†	−22.10†	0†	0†
AFAP	2.585†	1.143 [1.054, 1.241]	−36.99†	−22.14†	0†	1.49†
ALAP+safety layer	1.007†	0.821 [0.770, 0.875]	−21.21†	−21.18†	3.0e-03†	0†
ALAP	1.572†	0.927 [0.865, 0.992]	−26.87†	−21.17†	0†	0.57†
Round Robin+safety layer	1.095†	0.890 [0.834, 0.949]	−22.10†	−22.10†	0†	0†
Round Robin	1.095†	0.890 [0.831, 0.950]	−22.10†	−22.10†	8.7e-05†	0†
eMPC-HiGHS+ safety layer	1.145†	0.631 [0.583, 0.682]	−22.60†	−18.46†	0.41†	0†
eMPC-HiGHS	108.138†	105.926 [102.198, 109.683]	−1092.53†	2.10†	109.46†	1.7e-04†
N = 20						
SA-Attn (ours)	0.973 (0.971–0.976)	0.719 [0.676, 0.766]	−45.25	−43.21	0.20	0
Attention+penalty	1.569 (1.315–1.823)	0.748 [0.707, 0.793]	−57.16	−42.37	1.00	0.48
Attention+ PPO-Lagrangian	1.258 (1.076–1.440)	0.912 [0.862, 0.967]	−50.94	−47.25	6.6e-04	0.37
Attention+safety layer without departure bound	0.776 (0.709–0.843)	0.590 [0.555, 0.628]	−41.29	−40.58	0.07	0
Deep Sets+safety layer	0.722 (0.687–0.758)	0.527 [0.497, 0.559]	−40.22	−39.20	0.10	0
MLP+safety layer	0.894 (0.875–0.913)	0.706 [0.666, 0.747]	−43.66	−43.66	6.6e-04	0
MLP+penalty	1.795 (1.795–1.795)	1.009 [0.952, 1.075]	−61.68	−47.53	0.97	0.44
LLF+safety layer	1.099†	0.892 [0.846, 0.946]	−47.75†	−47.75†	0†	0†
LLF	1.099†	0.892 [0.842, 0.946]	−47.75†	−47.75†	0†	0†
sLLF+safety layer	1.099†	0.892 [0.843, 0.940]	−47.75†	−47.75†	0†	0†
sLLF	1.099†	0.892 [0.845, 0.944]	−47.75†	−47.75†	0†	0†
Price-threshold+ safety layer	0.788†	0.627 [0.591, 0.669]	−41.54†	−41.15†	0.04†	0†
Price-threshold	159.518†	153.133 [147.107, 159.274]	−3216.13†	43.35†	317.44†	8.51†
AFAP+safety layer	1.099†	0.891 [0.845, 0.943]	−47.75†	−47.75†	0†	0†
AFAP	1.518†	0.984 [0.920, 1.048]	−56.13†	−47.77†	0†	0.84†
ALAP+safety layer	1.028†	0.839 [0.795, 0.885]	−46.34†	−46.34†	1.8e-05†	0†
ALAP	1.091†	0.864 [0.817, 0.914]	−47.59†	−46.33†	0†	0.13†
Round Robin+safety layer	1.099†	0.892 [0.841, 0.945]	−47.75†	−47.75†	0†	0†
Round Robin	1.099†	0.892 [0.843, 0.943]	−47.75†	−47.75†	0†	0†
eMPC-HiGHS+ safety layer	0.846†	0.631 [0.592, 0.675]	−42.70†	−41.00†	0.17†	0†
eMPC-HiGHS	103.765†	102.828 [99.835, 105.746]	−2101.09†	0.97†	210.21†	1.9e-04†

Values: mean of the seed-averaged policy over all test episodes; a range in parentheses is the min–max over the 2 training seeds; † deterministic method (single run). Ranges for all columns, bootstrap confidence intervals and per-seed values are provided with the result data.
Optimality gap: J^{pen} of the soft-penalty perfect-information optimum minus J^{pen} of the method, divided by N (€ per charger and episode).
IQM: interquartile mean of the per-episode optimality gap; bootstrap CI over test replays. The zero-padded MLP is evaluated only for N ≤ 20.

6.3 Constraint satisfaction

Figure 4 shows energy not served, overload energy and E4 against N for the main methods. No transformer overload occurred in any of the 102,750 test episodes run through the safety layer (Table S8).

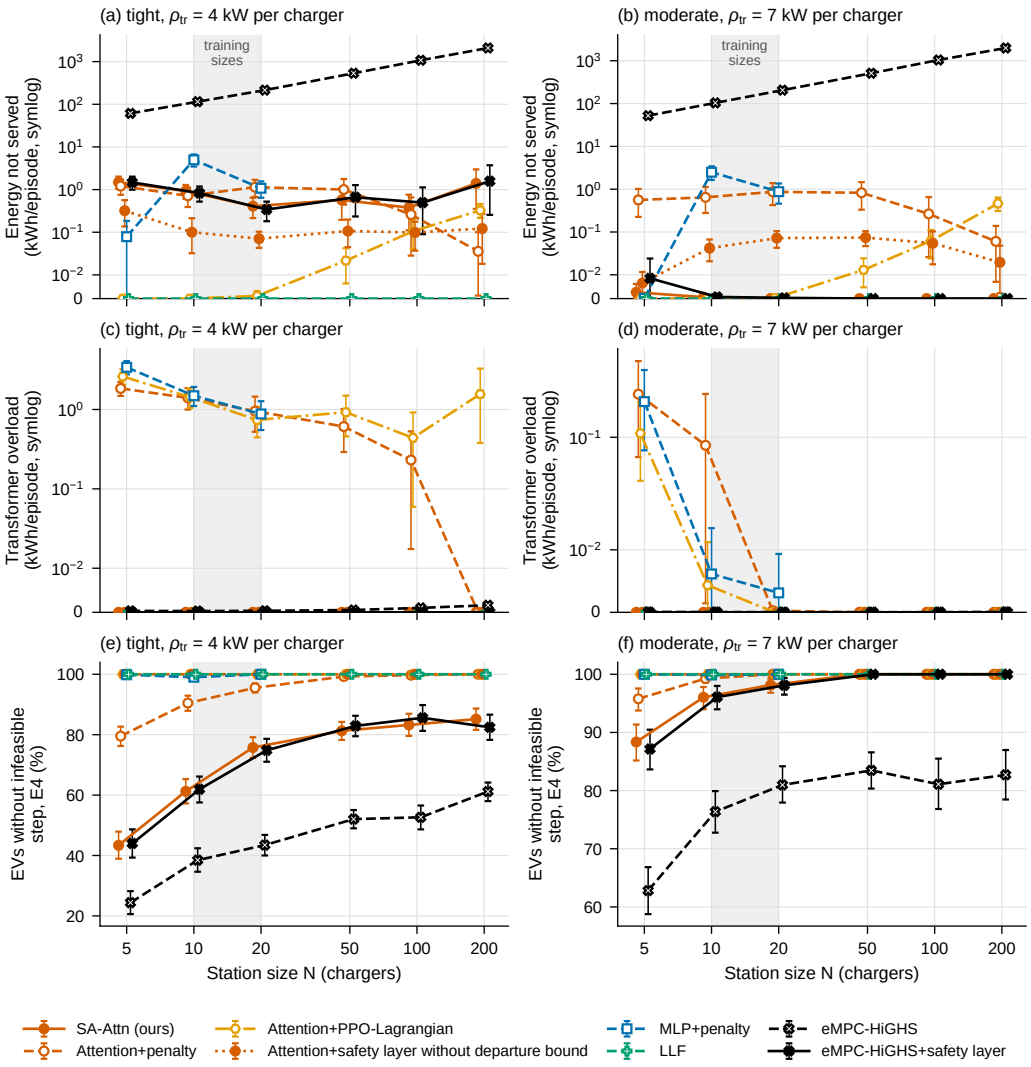


Figure 4: Constraint satisfaction across station sizes for both transformer ratings: energy not served, transformer overload energy and the share of EVs whose stay contains no infeasible step (E4), with 95% bootstrap intervals.

Without the layer, Attention+penalty had 3,730 overload steps, Attention+PPO-Lagrangian 3,274 and MLP+penalty 3,648 over all configurations. Averaged over the 36 test-grid configurations, 6.7%, 7.9% and 17.0% of their episodes had overload (Table S6). For every non-conformal method with the departure bound, no EV was left short in an episode without infeasible steps, consistent with Theorem 1(b). In Table S7, every fallback activation belonged to the case of excess required power, and the three cases that A6 excludes did not occur.

Learned policies with the departure bound caused infeasible steps. Over all episodes, SA-Attn had 17,678 infeasible steps and 39,771 EVs whose stay contained one, and it left 6,158 of 233,016 admitted EVs unserved. Deep Sets+safety layer had 12,122 infeasible steps, 27,949 affected EVs and 5,038 unserved EVs. LLF with the safety layer had one infeasible step and Price-threshold+safety layer 1,675. E4 of SA-Attn rose from 65.86% at $N = 5$ to 92.58% at $N = 200$, and infeasible steps per episode fell from 4.64 to 0.58. For Deep Sets+safety layer, E4 was higher at every size, with Holm-adjusted $p \leq 0.0043$ up to $N = 100$. At $N = 200$, its interval of $[0.9, 3.6]$ percentage points excluded zero, but the Holm-adjusted p was 0.94, so this size is inconclusive. The measured shortfall of SA-Attn was 0.009 times the a-priori deficit bound, as a mean over configurations of the episode means for episodes with a positive deficit.

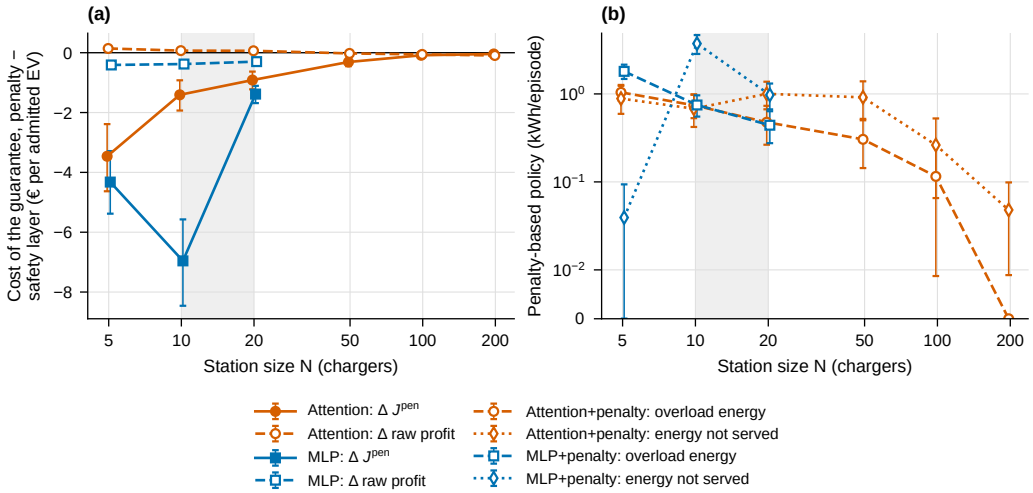


Figure 5: Cost of the guarantee across station sizes. (a) Penalty-based policy minus the policy with the safety layer, in J^{pen} and in raw profit per admitted EV, with 95% bootstrap intervals; a positive value means that the penalty-based policy scored higher. (b) Overload energy and energy not served of the penalty-based policies per episode.

6.4 Effect of the departure bound

The departure lower bound did not improve service for the learned attention policy. Attention+safety layer without departure bound had a lower mean gap than SA-Attn at every station size (Table S5). Its advantage ranged from 0.117 $[0.094, 0.143]$ at $N = 100$ to 1.266 $[0.927, 1.652]$ at $N = 5$, with Holm $p \leq 1.3 \times 10^{-30}$. The variant left 0.99% of the admitted EVs unserved against 2.24% for SA-Attn, with 0.09 against 0.43 kWh of energy not served per episode (Table S6). Summed over all episodes, these shares correspond to 1,551 against 6,158 unserved EVs (Table S8). At the stress level and $N = 20$, the variant left 2.59% of the EVs unserved against 12.04% (Table S15). The comparison is not a clean ablation, because the variant learned from the penalty reward with its own tuned weight, whereas SA-Attn learned from the profit reward.

SA-Attn also left more EVs unserved than its penalty-trained twin without any layer, 2.24% against 1.86% and 6,158 against 1,451. Its mean energy not served was lower, however, with 0.43 against 0.63 kWh per episode. Attention+PPO-Lagrangian left 0.20% of the admitted EVs unserved, the penalty-trained policy with the layer at deployment 0.46% and Price-threshold+safety layer 0.44%. LLF, sLLF, AFAP and round robin with the layer left no EV unserved. SA-Attn commanded exactly the departure lower bound on 27.8% to 42.3% of the occupied charger-steps, depending on N (Table S7).

6.5 Cost of the guarantee

Figure 5 and Table S10 compare SA-Attn with its penalty-trained twin on the same episodes. In J^{pen}, Attention+penalty minus SA-Attn was -3.462 $[-4.639, -2.388]$ € per admitted EV at $N = 5$ and approached zero with growing N, to -0.042 $[-0.094, 0.025]$ at $N = 200$. The Wilcoxon tests on the replay-level J^{pen} differences gave Holm-adjusted $p \leq 2 \times 10^{-4}$ at every size. At $N = 200$, the interval of the mean includes zero, whereas the median shift favours SA-Attn (Holm $p = 1.7 \times 10^{-6}$), so the size of this difference is inconclusive. In raw profit, the difference was positive at $N \leq 20$, from 0.142 $[0.075, 0.218]$ € per admitted EV at $N = 5$ to 0.064 $[0.039, 0.091]$ at $N = 20$. It was negative at $N \geq 50$, down to -0.092 $[-0.120, -0.065]$ at $N = 200$. For the MLP pair, the J^{pen} difference was between -6.958 and -1.383 € per admitted EV.

6.6 Further results

The conformal variant achieved coverages of 0.992, 0.950, 0.902 and 0.799 for targets of 0.99, 0.95, 0.90 and 0.80 (Table S12). Its gap was 27.341 at $\alpha = 0.01$, 1.292 at $\alpha = 0.05$ and 1.958 at $\alpha = 0.2$, against

1.210 with known departure steps. At $\alpha = 0.01$, the re-certified requests reduced the charged energy to 791.4 kWh per episode against 1396.5 kWh with known departures. With the safety layer added at deployment only, the penalty-trained policy had a gap of 0.930 at $N = 200$ (Table S5). Its difference to SA-Attn, -0.028 $[-0.061, 0.014]$ with Holm $p = 9.9 \times 10^{-6}$, is also inconclusive. One control step of SA-Attn took 7.76 ms at $N = 200$, against 5.99 ms for Deep Sets+safety layer (Table S13). On EV2Gym's 25-charger configuration, SA-Attn and Price-threshold+safety layer both had a gap of 0.507 (Table S16).

7 Discussion

[TODO: DRAFT. Under the SAUCIS AI policy, the interpretation of the results must be written by the authors. The authors rewrite this section in their own words and check every statement against the results; the draft below only proposes points to consider].

7.1 Interpretation

The safety layer delivered its two-layer result in every test configuration. Without exception, the transformer limit held in all episodes run through the layer, whereas penalty-based and Lagrangian training did not remove violations, with 3,730 and 3,274 overload steps. For every non-conformal method with the departure bound, the departure part held in the episode-level check, so the remaining shortfall came only from episodes with infeasible steps. In contrast, the variant without the departure bound left EVs unserved in episodes without any infeasible step.

The conditional part matters in practice, because the learned policies caused many infeasible steps. A plausible reading is that a profit-seeking policy keeps many EVs on their latest-start trajectory, so that the attention policy commanded exactly the lower bound on 28% to 42% of the occupied charger-steps. When the transformer then binds, these bounds cannot all be met at once. Rules that charge early rarely reached such states, and the penalty-trained policy with the layer at deployment had fewer infeasible steps. A lower bound that anticipates the aggregate capacity, rather than only each EV's own reachability, is a natural next step.

The conditional guarantee therefore had a price for the learned attention policy. With the bound, it left more EVs unserved than the variant without the bound, its penalty-trained twin and the Lagrangian policy. That variant also had a lower gap at every size. Part of this difference may come from training, because that variant learned from the penalty reward and not from the profit reward. Rules that charge early left no EV unserved with the same bound. The departure part is thus most useful with controllers that do not ride their lower bounds, and a profit-seeking learner can turn it into a source of infeasible steps.

Zero-shot transfer worked in the sense tested here. From 50 to 200 chargers, the gap of the attention policy stayed within 0.04 € per charger of a policy trained at 50 chargers, with intervals up to 0.05. Only the smallest station was clearly harder.

The architecture comparison does not support an advantage of attention. Instead, the Deep Sets policy had lower gaps and fewer infeasible steps than the attention policy at every station size of the test grid, at lower inference cost. One possible reason is that the safety layer already performs the coupling between chargers, so that a shared per-token policy with a pooled context suffices. Two seeds and the limited tuning budget, however, do not allow a firm ranking of the two architectures.

The comparison with rules shows that the safety layer is a reusable component. A price-threshold rule with the layer had a lower mean gap than the attention policy at every station size of the test grid, and the Deep Sets policy beat it only from 20 chargers upward. At the training size of 10 chargers, every rule with the layer had a lower mean gap than the attention policy. The experiments therefore show no benefit of the attention policy over a simple price-aware rule with the same layer, under the stated tuning budget of 12 configurations per method.

In penalised profit, the penalty-trained twin scored lower at every size, with an interval including zero only at 200 chargers, where the Wilcoxon test still favoured the policy with the layer. Measured in raw profit, the twin earned at most about 0.14 € more per admitted EV at small stations and less at large

ones. For the conformal bound, the achieved coverage stayed within 0.002 of each target and fell below it only at 0.80, but $\alpha = 0.01$ cut the admitted requests so strongly that service collapsed. Levels of 0.05 and 0.1 stayed close to known departures in gap, but their shortfall was six to seven times as large.

[TODO: Practical implications for charge point operators, to be written by the authors. Possible points: value of a transformer guarantee for every raw action, the conditional departure result, the conformal level and inference time].

7.2 Limitations

Each learned method was trained with two seeds, so between-seed variability is shown only as a range, and the tuning budget was the planned 12 configurations per method. The planned ablations, penalty-weight and other sensitivity studies, further MPC variants, the EV-GNN baseline and an example-day analysis were not run, and the only sensitivity is the stress rating. Our departure check is at episode level, because the logs do not record per EV whether it always received its lower bound.

The departure result rests on declared departure times and requirements (A2), and the conformal extension gives only marginal coverage. On the theory side, the transformer result needs the one-step model A3 and the headroom of A6, and a real battery management system may draw less than commanded. EV2Gym could not build some training days, because negative residential loads produced a negative noise scale. Training seeds were therefore redrawn, once per tuning job and twice per training job, whereas all test replays were built unchanged. The hard optimum was sensitive to the shortfall tolerance on a few replays, and 3 of 3,550 replays needed additional solver fallbacks, primal simplex or a tighter shortfall tolerance.

The experiments use EV2Gym's lossless accounting with equal charging and discharge prices, so profits from V2G arbitrage may be optimistic. All results come from simulation with synthetic arrivals and Dutch day-ahead prices, and grid quantities scale linearly with N. Our eMPC-HiGHS uses exact prices and arrivals inside its horizon, and the MLP is reported only up to 20 chargers. The optima ignore the information limits of an online controller, so part of every gap is the price of not knowing the future.

8 Conclusion

This paper combined size-agnostic set policies with a feasibility-preserving safety layer for V2G charging control at a CPO station. The layer projects raw actions onto taper-aware per-EV bounds and the two-sided transformer limit in $O(N \log N)$ time, during training and deployment. Theorem 1 gives a two-layer result. Under the one-step model and the grid-headroom condition, the transformer limit holds at every step for every sequence of raw actions, and every EV whose stay contains no infeasible step meets its admitted departure requirement.

In EV2Gym, no transformer overload occurred in any episode run through the safety layer. For every non-conformal method with the known-departure bound, no EV was left short in an episode without infeasible steps. Attention and Deep Sets policies trained at 10 and 20 chargers transferred to 200 chargers. The attention policy stayed within 0.04 € per charger and episode of a policy trained at 50 chargers in the point estimates. As an architecture, attention showed no advantage over Deep Sets. A price-threshold rule with the same layer had a lower mean gap than the attention policy at every station size of the test grid and than the Deep Sets policy below 20 chargers.

Learned policies caused infeasible steps, and 7.4% to 34.1% of the admitted EVs of the attention policy had one in their stay, depending on station size. With the departure bound, the attention policy left more EVs unserved than a variant without the bound and than its penalty-trained twin. In penalised profit, it scored higher than this twin, with an inconclusive difference at 200 chargers. [TODO: The authors confirm or rewrite these conclusions after rewriting the Discussion].

For a CPO, three limitations matter most. First, the departure result needs declared requirements and departure times, or a calibrated departure bound with marginal coverage only. Second, the transformer result assumes that each EV draws the commanded power, and real battery management systems may draw less. Third, the evidence comes from simulation with two training seeds per learned method and lossless economics. Future work can add a capacity-aware lower bound that reduces infeasible steps,

measured-energy feedback for hardware and several transformers with linearised voltage limits. Tests with recorded charging sessions would narrow the remaining gap between the model and practice.

Article Information Form

Authors' Contributions
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Conflict of Interest
The authors declare that there is no conflict of interest regarding the publication of this paper. [TODO: Confirm this declaration].

Support/Supporting Organizations
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Ethical Approval
[TODO: Confirm the following wording]. This study is a simulation study that involves no human participants, no human data collected by the authors and no animals, so ethics committee approval was not required. The authors declare that they complied with research and publication ethics.

Availability
The code, the scenario configurations, the test replays and the analysis scripts will be released in a public repository. The link is withheld for double-blind review. The experiments use the public EV2Gym code at commit `55ef368c` without changes to its source.

Artificial Intelligence Statement
[TODO: To be written by the authors. A factual record of the AI contributions is kept in `review/ai_contribution_record.md`].

Plagiarism Statement
This article has been scanned by iThenticate™.

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